What Missionary Control Left in the Language of Two 1880s isiXhosa Newspapers

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Abstract

From 1884 the missionary *Isigidimi sama-Xosa* and the African-owned *Imvo Zabantsundu* ran side by side. Did missionary control show in *what* they printed or *how* they used words? Imvo was not more political in vocabulary: of ten political concepts fixed in advance, only parliament differs (3.3×, uncorrected). Exploratory religion differs most: Isigidimi uses it four times as often, in readers' letters too, and the gap holds both in the same years and under the same editor, Jabavu. Framing evidence is weak and not specific to political words. Old spelling penalises only the African-language tokenizer. The results are consistent with control shaping what the papers printed; whether it shaped how they wrote lies below our detection floor.

Introduction and Background

Where did missionary control leave its trace in the language of the earliest isiXhosa press? In the IsiXhosa Intellectual Traditions archive (Schoots et al., 2023a), *Isigidimi sama-Xosa* was a missionary paper; *Imvo Zabantsundu*, founded in 1884 by Isigidimi's former editor J. T. Jabavu, was the first African-owned isiXhosa paper, free "from any missionary control" (Schoots et al., 2023a, p. 55). From late 1884 both had African editors, Gqoba and Jabavu (Schoots, 2023b, p. 71). Comparing the papers contrasts control, but also two editors, places and audiences, which one paper per side cannot separate. So we also hold the editor constant: Jabavu edited Isigidimi in 1881–84, then Imvo.

Communication research separates *agenda-setting*, what a paper covers (McCombs & Shaw, 1972), from *framing*, how it presents it (Entman, 1993). We measure the first as **salience**, how often a topic's words appear, and the second with word embeddings (Hamilton et al., 2016a). Such studies mostly use modern news (Card et al., 2015; Field et al., 2018); Schoots (2023b) counted root words in these papers. We add controls that make both testable, the same-editor contrast, and a letters-versus-rest split to ask *how* control worked; finally, we ask whether this history still shapes how machines read the archive (Petrov et al., 2023; Rust et al., 2021). **H1**: the papers differ in how often they use political and religious words. **H2**: Isigidimi's political words sit closer to religious contexts. **H3**: historical text and its spelling raise tokenizer cost.

Methods

Data. We use the archive's human-corrected transcriptions, not spell-checked (Schoots et al., 2023a, p. 59), from November 1884 to 1888: 286,663 tokens of Isigidimi and 1.31M of Imvo.

Salience. Concepts are word stems adapted from Schoots (2023b) (Table S1): ten political concepts fixed in advance, four neutral controls, and religion (*tixo*, *kristu*). Religion is exploratory: the archive's authors expect it (Schoots et al., 2023a, p. 57), but a classifier trained on the same years chose it (logistic regression, issue-grouped nested cross-validation, 0.83 balanced accuracy; chance 0.50). Ratios carry issue-bootstrap CIs; Jabavu's Isigidimi runs from January 1881 to October 1884.

Threat	Response
Different years	Nov 1884–1888 only; month-matched draws
Imvo 4.6× larger	Ten Imvo samples of Isigidimi's size
Frequency effects	2000 matched random word sets (Monroe et al., 2008)
Change over time	Each paper's own drift in other years
Unstable embeddings	Tune for seed stability; average ten runs
Non-political effects	Four neutral control concepts
One paper per side	Same editor in both; place, audience unfixed

Table 1: Threats to a fair comparison, and our response.

Letters. A letter opens by addressing the editor (*nkosi mhleli … kaundifakele*, 'sir, editor … please print') and ends with a signature; 28 of 30 random detections were genuine letters (Table S5).

Framing. Embedding shift is a weak proxy for Entman's framing: it also reflects sense, grammar and topic. We train fastText embeddings (Bojanowski et al., 2017), tuning 34 settings on other years for seed stability, since unstable embeddings fake shifts (Antoniak & Mimno, 2018) (0.83; word2vec 0.64; Table S8). This favours a grammar-heavy window of 1, so we also report windows 2 and 5. A word's *profile shift* is $1 - r$ between its similarities to 500 common words in each paper (Hamilton et al., 2016b); chance comes from shuffling issues (Dubossarsky et al., 2017). The directional test, aimed at framing, asks whether political words sit nearer religious or political anchors (Table S4); to read it, we count religious and legal words near *umhlaba*.

Tokenizers. We count subwords for XLM-R (Conneau et al., 2020), Serengeti (Adebara et al., 2023) and mBERT (Devlin et al., 2019) on modern news (Adelani et al., 2023), on the same news in old spelling (per word) and on our text (per character).

Results

H1: different things, but not more politics. Of ten political concepts, only parliament differs: Imvo uses it 3.3 times as often ($p = 0.018$ against matched sets; Fig. 1); law, vote, rights, land and government do not. Exploratory religion does: Isigidimi uses it 4.2 times as often ($p = 0.019$). Neither survives correction for 15 tests ($p < 0.003$). (CIs ask if rates differ; matched sets, if a concept differs more than random words.) The religious gap holds in letters (0.33) as in the rest (0.22; Fig. 1, right). Under the same editor, Jabavu, Imvo carried religious words a third as often as Isigidimi (0.34,

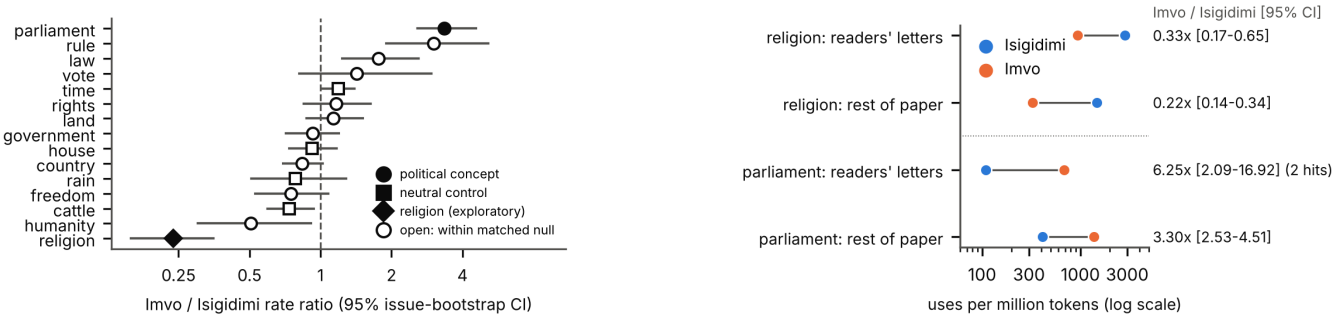


Figure 1: What the papers printed. Left: Imvo/Isigidimi rate ratio (95% CI; filled: $p < 0.05$ against matched sets, uncorrected). Right: the religious gap holds in readers' letters too.

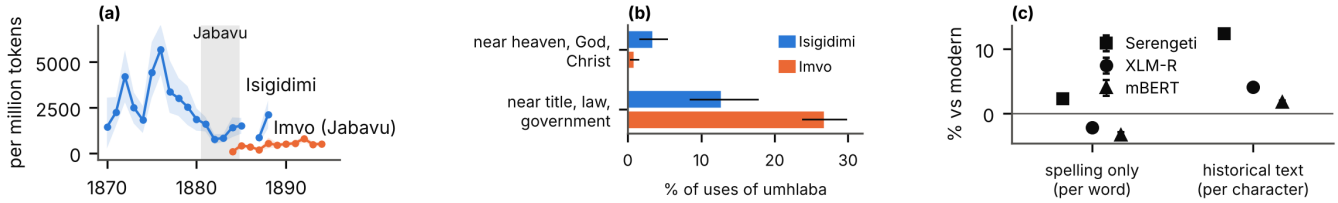


Figure 2: (a) Religious words per million tokens; grey: Jabavu edits Isigidimi, then Imvo; Gqoba edits Isigidimi from late 1884. (b) Uses of *umhlaba* near religious or legal words (95% CI). (c) Extra subwords against modern text (95% CIs smaller than markers).

CI 0.24–0.47, matched-set $p = 0.028$), and parliamentary words 1.7 times as often ($p = 0.082$). Under Gqoba, Isigidimi carried religious words 1.45 times as often as under Jabavu ($p = 0.130$). Within Isigidimi, religious words had already fallen to 0.31 of their 1876–80 level by 1881–84, while parliament rose 50-fold and vote 12-fold ($p \leq 0.004$; Fig. 2a; Table S7).

H2: framing differences are weak and not specific. Word use differs overall (profile shift 0.348 vs 0.230 by chance); no concept stands out: parliament reaches the 96th percentile of matched sets, the control *cattle* 90th–99th (Fig. S1). Political words lean towards religious anchors in Isigidimi in all five settings (Δ +0.44 to +0.76 z-units; Bonferroni-significant in three; Table S2). But without *land* the lean is rarely significant ($p = 0.010$–0.256); the control *house* leans further (+0.84 to +1.51); dropping the anchor *endlu* ('in the house') changes neither; and *umhlaba* sits near religious words in 3.4% of Isigidimi's uses but 0.9% of Imvo's (Fig. 2b). A one-word concept must shift by 0.15 (1.645 matched-set SDs) to be detected, more than the overall gap (0.118).

H3: historical text costs more. Per character, our text needs 12.4%, 4.1% and 1.7% more subwords with Serengeti, XLM-R and mBERT; old spelling alone raises only Serengeti's (+2.2% per word; Fig. 2c).

Discussion

Is it control, or time? (H1). Religious words were falling in Isigidimi anyway, so we triangulate over a 2×2 of control and editor with one cell missing. Imvo vs Gqoba's Isigidimi fixes the years (0.24); Jabavu's two papers fix the editor (0.34). Both show the gap: two contrasts with different confounds agree in direction (their p-values are uncorrected). Jabavu's Isigidimi predates the years in which the classifier chose religion, so this contrast partly replicates it on held-out text. Were the gap only the trend, Isigidimi would have kept falling after 1884; it did not (1.45). A Jabavu-era Imvo as religious as his Isigidimi would have refuted control, and a further fall under Gqoba would have undermined it; neither occurred. Politics instead followed the times, rising within Isigidimi before Imvo existed. Whether Jabavu was right to leave is beyond our data; leaving brought less religion, not clearly more politics. The gap reached beyond editors and is not only a matter of which sections each paper ran: Isigidimi's letter-writers used religious words three times as often as Imvo's, so the papers drew on, or chose, different writers. One wrote that *ilizwi alizanga ukuba silipate, lize ukuba lisipate* ('the Word did not come for us to carry it; it came to carry us', or 'rule us'; 1 Nov. 1884; our gloss, for speaker review).

What the embeddings see (H2). The framing signal looks like a difference in meaning, not slant: Isigidimi more often uses *umhlaba* for the earth, Imvo for land under law, which alone would move the word in each model. The detection floor makes the null a bound: large framing differences are ruled out, small ones are not.

Responsible NLP (H3). History also reaches today's tools: written isiXhosa took shape in the mission press (cf. Gilmour, 2007), and its old spelling penalises only the African-language model, perhaps because Serengeti's vocabulary has twice XLM-R's share of *th*/*kh*/*ph* pieces (1.9% vs 0.9%); a one-model hypothesis. A normalised copy must not replace the source: the mapping is lossy (old *t* may be *t* or *th*). Our results describe two papers, not isiXhosa speakers; cultural readings without speaker review would be extractive (Bird, 2020).

Limitations and conclusion. Two papers, one window, a missing year, an incomplete 2×2, a heuristic letter split, a domain mismatch, unreviewed glosses (Table S1). The evidence fits control shaping what the papers printed, letters included; effects on political language lie below our detection floor.

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